SUTD's Double Diamond Design Thinking Pedagogy

Research summary, November 2026. Sources are listed at the end. Each claim names its source.

1. Summary

- SUTD teaches design thinking through its own **Design Innovation (DI)** methodology. DI uses the **4D double diamond**: Discover, Define, Develop, Deliver.
- DI is "built, in part, on the UK Design Council's 4Ds". SUTD adds four **mindsets**, 24 **principles**, a set of **methods** for each phase, and a blend of **Design Thinking** and **Systems Thinking** [1][2].
- SUTD treats the order of the phases as a guide, not a rule. One pass through the four phases is a "sprint". A real project has "multiple sprints, pivots, and leaps" between phases [1].
- All first-year (Freshmore) students learn the double diamond in **3.007 Design Thinking and Innovation**, which has more than 400 students in each cohort [3][4][5].
- From the AY2026 cohort, the Freshmore course becomes **Innovating with Design & AI (iDeA)**. Students use AI in each stage of the design process [6][7].
- SUTD also uses the double diamond outside its degree programmes: in youth programmes, teacher training and research on AI planning tools [8][9][10].

2. The double diamond at SUTD: the DI process

2.1 Origin

The UK Design Council published the Double Diamond in 2005. It shows two cycles of divergent and convergent thinking [11]. The first diamond is about the problem. The second diamond is about the solution.

SUTD and the SUTD-MIT International Design Centre (IDC) built the DI methodology on this model. The *Design Innovation Methodology Handbook* (1st edition, June 2021) gives the full method. Its authors are Carlye Lauff, Wee Yu Hui, Kenneth Teo, Sabrina Png, Amanda Swee, Arianne Collopy, Brandon Vargas and Kristin L. Wood [1]. A 2nd edition is cited in later SUTD papers [10].

2.2 The four pillars

The handbook defines DI as "a human-centred and inter-disciplinary methodology to innovate on and address complex challenges" [1]. DI has four pillars:

Pillar	Meaning
People	Start with the stakeholders and users, and with the internal team.
Process	The 4D double diamond. It balances divergent and convergent thinking.
Methods	Guided activities for each phase, for example interviews or a Pugh matrix.
Principles	Best practices that guide how people think, communicate and decide.

DI applies to Products, Services and Complex Systems ("PSS") [1].

2.3 The four phases

The handbook gives each phase a goal and a set of guiding questions [1]. SUTD's STEM-Unity

programme gives a short version for students [8].

Phase	Diamond	Goal (handbook)	Typical questions (handbook)
Discover	1: problem, diverge	Find and understand opportunities and needs with the stakeholders.	Who are the users and stakeholders? What are their needs? How do they behave and feel? What extreme conditions can inform us?
Define	1: problem, converge	Interpret and reframe needs. Map them into activities, functions and representations.	What are the users' journeys and emotions? What are the key insights? What system functions are necessary?
Develop	2: solution, diverge	Generate and model concepts from the opportunities.	How do we generate many ideas? What analogies can we use? Which 3 to 6 ideas come out of this sprint?
Deliver	2: solution, converge	Prototype and test concepts with users, again and again.	What questions must the prototype answer? How do we engage users? How do we mix quantitative and qualitative results?

The **Opportunity Statement** connects the two diamonds. It is the output of Define and the input to Develop [1].

2.4 The four mindsets

The handbook links four mindsets to the process [1]:

- **Empathy:** Understand and share the feelings of others, without judgement.
- **Mindfulness:** Pay attention to the present moment, on purpose, without judgement.
- **Joy:** Happiness and delight. The handbook says that a playful team gets more creative ideas.
- **Non-attachment:** Do not hold tightly to "my ideas, my solutions, my prototypes". Work towards the team goal.

The handbook uses an iceberg model. A change in how people think changes the conversation first, then perception, and then measurable outcomes [1].

2.5 The 24 principles

DI has 12 Design Thinking principles and 12 Systems Thinking principles [1].

- **Design Thinking:** Creativity Throughout; Adaptive Pathways; Embrace Open Resources; Expressive Collaboration; Curiosity for Context; Also Can; Appetite for Ambiguity; Empathy for All; Make, Test, Learn, Repeat; Free Space for Blue Skies; Pride in Art, Art in Craft, Craft in Pride; Celebrate Quantitative and Qualitative.
- **Systems Thinking:** Collaborate; Achieve Balance; See through New Eyes; Maintain Adaptive Feedback Loops; Integrate Problems; Take an Adaptive Stance; Combine Courage with Humility; Think like a Gardener, not a Watchmaker; Meta-cognition; Zoom in and Zoom out; Identify and Use Patterns; Learn from Problems.

2.6 Methods for each phase

The handbook calls the methods a "language of design". It puts each method in the phase where it fits best, and marks it as Design Thinking (DT) or Systems Thinking (ST). A team can use a method in other phases too [1].

Phase	Design Thinking methods	Systems Thinking methods

Discover	User Interviews, Site Analysis, Empathic Lead User, User Journey Map	Stakeholder Mapping, Influence Diagram, Benchmarking
Define	Affinity Analysis, Personas, Scenarios, Activity Diagram, Hierarchy of Purpose	Service/UX Blueprinting, House of Quality, System Architecture, Ishikawa Diagram, Systems Function Model
Develop	Brainstorming, Mind Mapping, C-Sketch (6-3-5), Design by Analogy, SWOT Analysis, Real? Win? Worth It?	Adjacency Diagram, SCAMPER, Pugh Matrix, Prioritisation Matrix
Deliver	Prototyping Canvas, Storyboarding, Mockups (Paper Prototypes), 2 x 2 Feedback Matrix, Pitching	Scaled Model, Immersive VR/AR, Desktop Walkthrough

Each method page gives what, why, time, materials, a procedure and a worked example [1].

2.7 Roles and tracking tools

- **DI Catalyst:** The handbook recommends one person in each project to act as a "DI Catalyst". This person suggests methods, makes sure each voice is heard, and has a "bias towards action". One rule: if a team talks about the same issue for more than 30 minutes, the team must stop and start to sketch [1].
- **Method Flow Chart and Design Signature:** These diagrams record the path that a real project took through the four phases, with the methods used and the time spent in each phase. The example project used 340 person-days. The handbook says that the order of the phases is "archetypal; it acts as a general guideline" [1].

3. How SUTD teaches the double diamond

3.1 Freshmore: 3.007 Design Thinking and Innovation

- **Who and when:** 3.007 is a Freshmore core course for all students, in Term 2, with 12 credits [3]. It is linked to **01.019 Design Thinking Project II**. The project series moves from a guided project (01.018, Term 1) to more self-guided work (01.020, Term 3) [12].
- **Structure:** The course site describes a double diamond sequence. Students first "discover and define a project scope" in a problem-centric phase. Then they "develop and deliver" ideas in a solution-centric phase. The process is iterative, with smaller cycles of divergence and convergence inside each phase [4].
- **Delivery:** The course uses e-lecture videos (released one week before), a one-hour cohort dialogue, a skills session and two cohort sessions of 2.5 hours [4].
- **Stance:** The course does not teach one correct way to design. It shows students "many different ways in which one can design" [4]. Mixed teams of instructors from different professions guide the student teams [3].
- **Large-cohort teaching:** Kaur et al. (IEEE TALE 2023) describe the course for more than 400 first-year students. They identify five process components: course delivery format, assignment design, diverse mentorship, access to past projects and teamwork facilitation. These components help students to "balance divergence and convergence", to explore past boundaries and to work through project-based learning [5].

3.2 From AY2026: Innovating with Design & AI (iDeA)

SUTD states that it is the first university to put Design AI into all first-year courses. From the AY2026 cohort, Freshmore has three terms of **Innovating with Design & AI** (course codes include 03.007A and 03.007B). Students use AI at each stage of the design process: user research, site and context analysis,

synthesis, ideation, concept development and prototyping [6][7]. SUTD's graduate programme IDeAL ("Innovating with Design & AI for Leaders") is also based on the iDeA framework and the "Double Diamond pedagogy" [7].

A Term 3 Freshmore elective, **02.005 Introduction to Human-Centred Design**, teaches empathy, ethics and context. It also introduces "intelligence-centred design" [13].

3.3 The wider idea: "Big-D" design

SUTD's design education uses the idea of **"Big-D" design**. Big-D includes architecture, products, software and systems, and the full value chain, with the humanities and social sciences [14].

4. Use outside the degree programmes

- **STEM-Unity (youth):** A six-week programme. In Stage A ("Learn & Ideate"), students learn the 4D double diamond and solve a community problem. In Stage B ("Build & Realise"), they build a prototype with community partners [8].
- **Teacher training:** SUTD Academy runs a one-day course for teachers. It covers the Design Council's Double Diamond, design briefs, a skills framework for design and assessment rubrics [9].
- **AI planning for teachers:** Liow, Khan and Ang (2025) built the IDPplanner, a GPT-based assistant based on DI. In a study with 33 teachers, AI-assisted project plans got higher scores for curriculum alignment, design thinking application, and coherence and flow. Teachers still adapted the plans to their own students [10].

5. Use in this workshop

The crew in this workshop follows the double diamond. Each agent works mainly in one phase:

Phase	Agent	Task
Discover	Deep researcher	Find evidence about the users and their problems.
Define	Design thinker	Write one problem statement and a "How might we" question (the Opportunity Statement).
Develop	Solutions engineer	Generate ideas and select one (like a Pugh matrix).
Develop / Deliver	Product designer	Design the user journey and the screens (storyboard, mockups).
Deliver	Marketing lead	Plan the launch and pitch the concept.

Ideas from the DI method to try with the crew:

- Use DI method names in the task descriptions, for example "make an affinity analysis" or "write a 2 x 2 feedback matrix".
- Run more than one sprint. Feed the canvas feedback back into a second run.
- Give one agent the "DI Catalyst" role. It checks that each phase has evidence before the crew continues.

6. Limits of this research

- The 3.007 course site (asd.courses.sutd.edu.sg) was not reachable during this research. Section

3.1 uses search-result extracts of that site [4].

- For the TALE 2023 paper, only the abstract was available [5].
- The iDeA course pages give no full syllabus yet. Its exact use of the double diamond is not confirmed [6][7].
- The handbook is copyright 2021 Kristin L. Wood, with all rights reserved. This summary quotes short parts only. Read the full handbook at the link in [1].

7. Sources

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